

Structured Query Language (SQL)

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Recap: Relational Algebra

- A query language of a set of operations that take one or two relations as input and produce a new relation as their result.
- Operators
 - select: σ
 - project: Π
 - union: $\cup$
 - set difference: $-$
 - Cartesian product: $\times$
 - Natural Join $\bowtie$
 - Theta join $\bowtie_{\theta}$
 - rename: ρ
- Assignment
- Expression tree

History: Structured Query Language (SQL)

- ANSI and ISO standard SQL:

- SQL-86
- SQL-89
- SQL-92 — *most used in commercial databases*
- SQL:1999 (language name became Y2K compliant!)
- SQL:2003

Data Definition Language (DDL)

- Define database schema

Example: **create table** *instructor* (

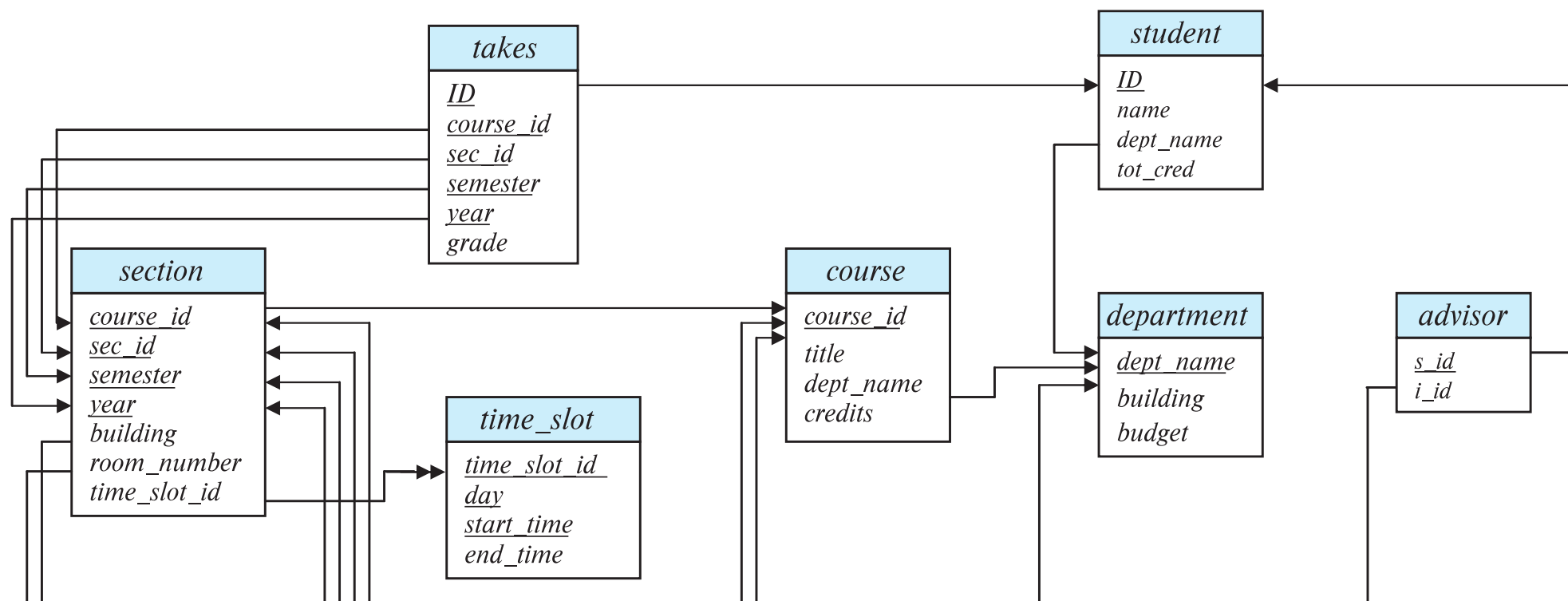
ID **char**(5),
name **varchar**(20),
dept_name **varchar**(20),
salary **numeric**(8,2))

小数点前 8位, 后 2位

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The *instructor* table

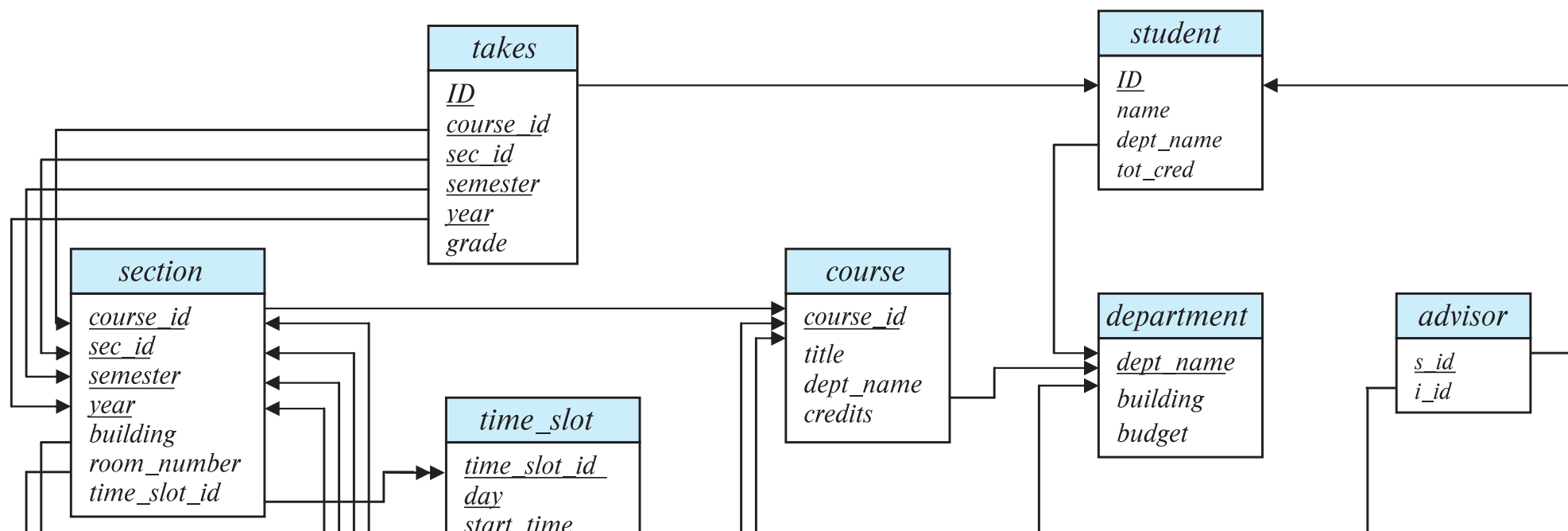
University Database: Create table for student



```

❑ create table student (
    ID          varchar(5),
    name        varchar(20) not null,
    dept_name   varchar(20),
    tot_cred    numeric(3,0),
    primary key (ID),
    foreign key (dept_name) references department);
    
```

University Database: Create table for takes



```

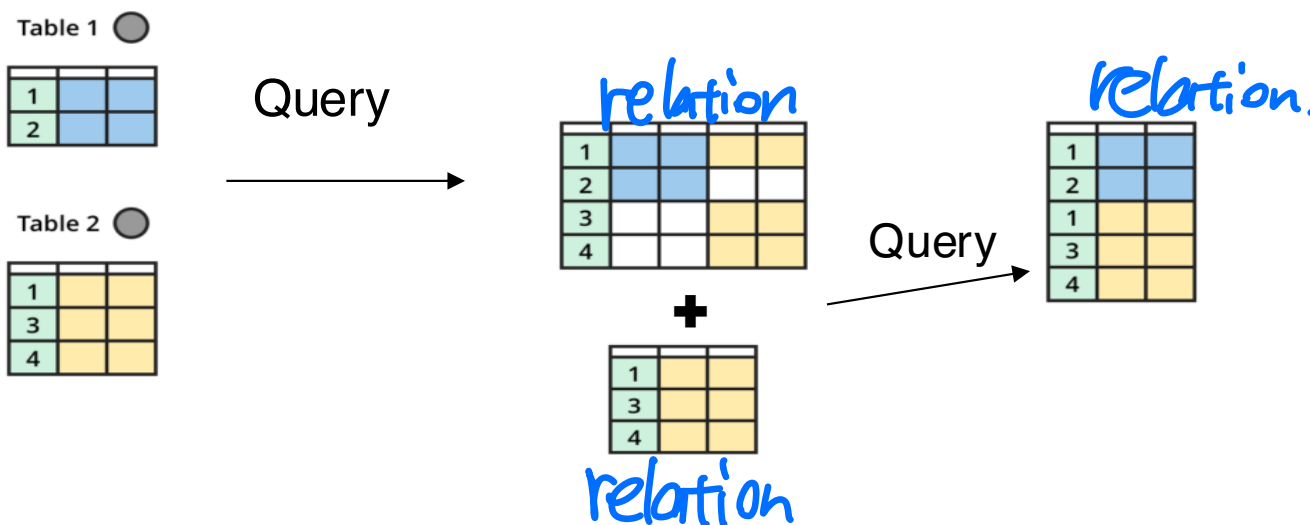
❑ create table takes (
    ID          varchar(5),
    course_id   varchar(8),
    sec_id      varchar(8),
    semester    varchar(6),
    year        numeric(4,0),
    grade       varchar(2),
    primary key (ID, course_id, sec_id, semester, year) ,
    foreign key (ID) references student,
    foreign key (course_id, sec_id, semester, year) references section);
  
```

1 primary key
multiple foreign keys

Data Manipulation Language (DML)

- Language for accessing and updating the data
- DML also known as **query language**
- Declarative**
- Query returns a relation $\Rightarrow$ compositional and close
- Compositional

access
update.



Basic Query Structure

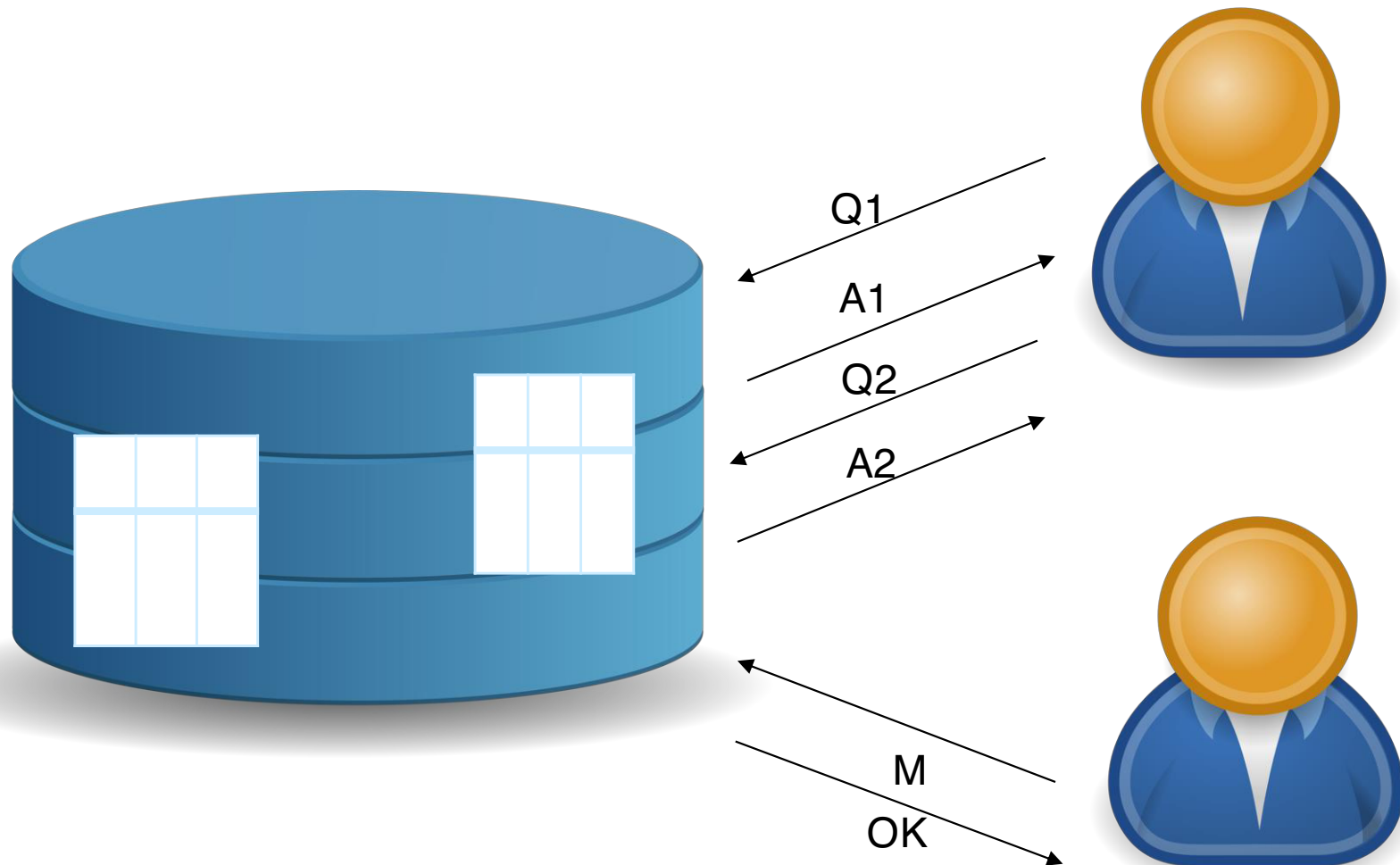
Clause 从句, 分句

- A typical SQL query has the form:

```
select  $A_1, A_2, \dots, A_n$ 
from  $r_1, r_2, \dots, r_m$ 
where  $P$ 
```

- A_i represents an attribute
- r_i represents a relation
- P is a predicate. condition
- The result of an SQL query is a relation.
- Update
 - **insert into** *instructor* **values** ('10211', 'Smith', 'Biology', 66000);
 - **delete from** *student*
 - **drop table** *student*
 - alter
 - **alter table** *student* **add** *age* *numeric(0, 150)*
 - **alter table** *student* **drop** *age*

Basic steps in creating and using relational DB



Wrap up

- Data Definition Language (DDL)
- Data Manipulation Language (DML)
- Basic Query Structure
- Basic steps in creating and using relational DB

duplication 重复

Demonstrations

select distinct xxx from xv
去重

concepts.

- Basic Query Structure of SQL Queries
- Additional Basic Operations
- Set Operations
- Null Values
- Aggregate Functions
- Nested Subqueries
- Modification of the Database

Schema relation
xxx.

可用数学运算符在 select 中

select $\frac{xxx}{a}$ from ...

* (every attribute)

select xxx from { xxx nature join xxx ;

= xxx, xxx where xxx = xxx;

select A.xxx from xxx as A, xxx as B where
rename.

(like).

select xxx from xxx where xxx like '%in%';

Demonstrations

select xxx from xxx

order by xxx;

任意
长度字符串

- Join Expressions

- Views

- Transactions

- Integrity Constraints

- SQL Data Types and Schemas

- Index Definition in SQL

- Authorization

descending order

xxx, xxx (can be) if same,
order by
other

order by xxx desc;

reverse order.

select xxx from xxx where a between xxx and xxx

(包含等号)

select xxx from xxx where ... union select xxx from ...

means or (交集)

intersect.

相交集 (子集)

Basic Query Structure (demo)

Using **All** to remain duplications (union all) except 去除

- Find the names of all instructors whose name includes the substring "in".
- String Operations
 - The operator **like** uses patterns that are described using two special characters:
 - percent (%). The % character matches any substring. 任意长度
 - underscore (_). The _ character matches any character 一位长度
- Find the names of all instructors whose name has 4 characters.

department	instructor	teaches	section	takes	student	course	advisor
<u>dept_name</u> building budget	<u>ID</u> name dept_name salary	<u>ID</u> <u>course_id</u> <u>sec_id</u> <u>semester</u> <u>year</u>	<u>course_id</u> <u>sec_id</u> <u>semester</u> <u>year</u> building room_number time_slot_id	<u>ID</u> <u>course_id</u> <u>sec_id</u> <u>semester</u> <u>year</u> grade	<u>ID</u> name dept_name tot_cred	<u>course_id</u> title dept_name credits	<u>s_id</u> i_id

FIN

Any questions?